

```
In [1]: import numpy as np
```

```
In [2]: a=np.array([[10,15,25],[35,14,12],[36,28,19]])  
print(a,type(a))
```

```
[[10 15 25]  
 [35 14 12]  
 [36 28 19]] <class 'numpy.ndarray'>
```

```
In [3]: maxv=np.amax(a)  
print("Max element=",maxv)
```

```
Max element= 36
```

```
In [4]: print(a)  
maxv=np.amax(a)  
colmax=np.amax(a,axis=0)  
rowmax=np.amax(a,axis=1)  
print("Max element=",maxv)  
print("Col Max Elements=",colmax)  
print("Row Max Elements=",rowmax)
```

```
[[10 15 25]  
 [35 14 12]  
 [36 28 19]]  
Max element= 36  
Col Max Elements= [36 28 25]  
Row Max Elements= [25 35 36]
```

```
In [5]: print(a)  
minv=np.amin(a)  
colmin=np.amin(a,axis=0)  
rowmin=np.amin(a,axis=1)  
print("Min element=",minv)  
print("Col Min Elements=",colmin)  
print("Row Min Elements=",rowmin)
```

```
[[10 15 25]  
 [35 14 12]  
 [36 28 19]]  
Min element= 10  
Col Min Elements= [10 14 12]  
Row Min Elements= [10 12 19]
```

```
In [7]: #cal mean of ndarray
print(a)
m=np.mean(a)
print("mean=",round(m,2))
```

```
[[10 15 25]
 [35 14 12]
 [36 28 19]]
mean= 21.56
```

```
In [8]: a=np.array([[2,1],[3,4]])
print(a)
```

```
[[2 1]
 [3 4]]
```

```
In [9]: m=np.mean(a)
colmean=np.mean(a,axis=0)
rowmean=np.mean(a,axis=1)
print("Mean=",m)
print("Col Mean=",colmean)
print("Row Mean=",rowmean)
```

```
Mean= 2.5
Col Mean= [2.5 2.5]
Row Mean= [1.5 3.5]
```

```
In [10]: #cal median of ndarray
a=np.array([[10,15,25],[35,14,12],[36,28,19]])
print(a,type(a))
```

```
[[10 15 25]
 [35 14 12]
 [36 28 19]] <class 'numpy.ndarray'>
```

```
In [11]: med=np.median(a)
colmed=np.median(a,axis=0)
rowmed=np.median(a,axis=1)
print("Median=",med)
print("Col Median=",colmed)
print("Row Median=",rowmed)
```

```
Median= 19.0
Col Median= [35. 15. 19.]
Row Median= [15. 14. 28.]
```

```
In [12]: a=np.array([[2,1],[3,4]])
print(a)
```

```
[[2 1]
 [3 4]]
```

```
In [13]: med=np.median(a)
colmed=np.median(a,axis=0)
rowmed=np.median(a,axis=1)
print("Median=",med)
print("Col Median=",colmed)
print("Row Median=",rowmed)
```

```
Median= 2.5
Col Median= [2.5 2.5]
Row Median= [1.5 3.5]
```

```
In [14]: #cal var of ndarray
a=np.array([[2,1],[3,4]])
print(a)
```

```
[[2 1]
 [3 4]]
```

```
In [15]: v=np.var(a)
colvar=np.var(a,axis=0)
rowvar=np.var(a,axis=1)
print("Var=",v)
print("Col Var=",colvar)
print("Row Var=",rowvar)
```

```
Var= 1.25
Col Var= [0.25 2.25]
Row Var= [0.25 0.25]
```

```
In [17]: #cal std of ndarray
a=np.array([[2,1],[3,4]])
print(a)
```

```
[[2 1]
 [3 4]]
```

```
In [18]: s=np.std(a)
colstd=np.std(a,axis=0)
rowstd=np.std(a,axis=1)
print("Std=",s)
print("Col Std=",colstd)
print("Row Std=",rowstd)
```

```
Std= 1.118033988749895
Col Std= [0.5 1.5]
Row Std= [0.5 0.5]
```

```
In [19]: m=np.mode(a) # AttributeError: module 'numpy' has no attribute 'mode'
```

```
-----
AttributeError                                Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_10176\3355755762.py in <module>
----> 1 m=np.mode(a)

~\anaconda3\lib\site-packages\numpy\__init__.py in __getattr__(attr)
    311             return Tester
    312
--> 313         raise AttributeError("module {!r} has no attribute "
    314                               "{!r}".format(__name__, attr))
    315

AttributeError: module 'numpy' has no attribute 'mode'
```

```
In [20]: #In Python Programming, we have statistics module and it contains mode() and mu
```

```
In [21]: import statistics as s
```

```
In [22]: lst=[10,20,30,10,20,40,50,60,70,10,20,10]
freq=s.mode(lst)
print("Frequently Repeated Element=",freq)
```

Frequently Repeated Element= 10

```
In [23]: lst=[10,20,30,10,20,40,50,60,70,10,20,10,20]
freq=s.mode(lst)
print("Frequently Repeated Element=",freq)
```

Frequently Repeated Element= 10

```
In [24]: lst=[20,10,30,10,20,40,50,60,70,10,20,10,20]
freq=s.mode(lst)
print("Frequently Repeated Element=",freq)
```

Frequently Repeated Element= 20

```
In [25]: lst=[20,10,30,10,20,40,50,60,70,10,20,10,20]
freq=s.multimode(lst)
print("Frequently Repeated Element=",freq)
```

Frequently Repeated Element= [20, 10]

```
In [26]: lst=[20,10,30,10,20,40,50,60,70,10,20,10,20]
a=np.array(lst)
freq=s.mode(a)
print("Frequently Repeated Element=",freq)
```

Frequently Repeated Element= 20

```
In [27]: lst=[20,10,30,10,20,40,50,60,70,10,20,10,20]
a=np.array(lst)
freq=s.multimode(a)
print("Frequently Repeated Element=",freq)
```

Frequently Repeated Element= [20, 10]

In []: